

Tourism Structural Explorer: Coordinated Views for Exploring Tourism Dependence, Employment Composition, and Development Indicators

S.P. Verbeek (2010498)

P.C. Villadangos (2108602)

ABSTRACT

This final report presents *Tourism Structural Explorer*, an interactive Dash dashboard for exploring how international tourism dependence relates to employment-sector composition and selected development indicators across countries and years. The system uses World Development Indicators data and coordinated views: association scatterplots, a small-multiple association overview, ternary employment-composition plots, a structural-shift dotplot, and a data-coverage matrix. Two task-based use cases show how users can move from cross-country association patterns to focus-country temporal pathways and data-reliability checks. The results show that tourism dependence and service employment do not follow a uniform pattern across the selected countries: Morocco, for example, is not among the highest-service countries in the 2020 snapshot, but shows the strongest service-sector increase over time. The analysis is exploratory: the dashboard supports comparison, outlier detection, temporal pathway inspection, and hypothesis generation, but it does not estimate causal effects.

Index Terms: Human-centered computing—Visualization—Visualization application domains; Human-centered computing—Visualization—Visualization systems and tools

1 INTRODUCTION

Tourism is often discussed as a development pathway because it can bring foreign exchange and service-sector demand. The relationship between tourism dependence and structural transformation is not simple: countries with high tourism receipts may also differ in GDP per capita, urbanization, poverty, inequality, region, and data availability. The goal of this project is therefore not to prove that tourism causes structural change. Instead, the goal is exploratory visual analysis of cross-country relationships, outliers, temporal employment pathways, and data-coverage limitations.

The intended users are development-oriented students, analysts, and policy researchers who need to compare countries rather than inspect one isolated statistic. Visualization is appropriate because the questions are open-ended and comparative. A single correlation coefficient cannot show whether regional groups cluster differently, whether a focus country follows a service-oriented pathway over time, or whether an apparent pattern is based on sparse data. Following Munzner’s What/Why/How framing [3], the dashboard links domain data and tasks to visual encodings and interactions that support comparison, selection, and details on demand.

2 RELATED WORK

Existing development-data platforms provide broad support for country-level indicator exploration. World Bank DataBank supports custom queries and visualizations such as tables, charts, and maps for time-series datasets, including World Development Indicators [12]. Gapminder provides interactive tools for comparing countries and indicators over time [2], while Our World in Data publishes interactive data visualizations and explanatory articles for long-run global development indicators [5]. These tools are valuable for broad exploration, but they are not designed around the specific workflow of linking tourism dependence to employment-sector composition, structural employment shifts, and explicit data-coverage checks.

Tourism-specific data sources such as the UN Tourism Data Dashboard and WTTC research focus on tourism indicators and economic impact. UN Tourism reports inbound and outbound tourism indicators such as arrivals and receipts at global, regional, and national levels [11]. WTTC publishes data and research on the economic, environmental, and social impact of travel and tourism across countries and regions [14]. In contrast, *Tourism Structural Explorer* uses tourism receipts as a dependence indicator and connects it to employment composition and development outcomes in a coordinated visual workflow.

The composition views are motivated by the fact that employment shares in agriculture, industry, and services form a three-part composition. Compositional data are commonly analyzed as relative parts of a whole [1], and ternary visualizations represent three components that sum to a constant in a triangular coordinate system [8]. Because ternary plots are less familiar than Cartesian charts, the dashboard combines them with hover details and a structural-shift dotplot.

3 DATA AND TASK ABSTRACTION

3.1 Domain Data and Derived Attributes

The project uses World Development Indicators data [13]. The raw variables include international tourism receipts as a percentage of total exports, employment shares in services, industry, and agriculture, employment group variants for male and female workers, GDP per capita, urban population share, poverty headcount, Gini index, country, year, and a project-assigned region category.

International tourism receipts as a percentage of total exports was selected as the main tourism-dependence indicator because it normalizes tourism revenue by a country’s export structure. This is more comparable across countries of different absolute economic size than raw tourism receipts or visitor arrivals. Alternative tourism indicators, such as international arrivals or absolute receipts, capture tourism volume but not necessarily dependence. The limitation is that this single indicator does not capture domestic tourism or indirect tourism employment, so the dashboard supports exploratory comparison rather than a complete causal analysis.

Several values are derived by the project rather than directly provided as raw WDI fields. Gender-average employment is computed as the arithmetic mean of the male and female employment shares for the same country, year, and sector:

$$v_{avg}(c, y, s) = \frac{v_{male}(c, y, s) + v_{female}(c, y, s)}{2}.$$

For country c , year y , and attribute a , the latest-available value is the most recent non-missing value at or before y :

$$v_{latest}(c, y, a) = v(c, t, a), \quad t = \max\{t' \leq y \mid v(c, t', a) \text{ is available}\}.$$

Data coverage is the share of selected years with non-missing values for a country and indicator:

$$coverage(c, a) = \frac{|\{y \mid v(c, y, a) \text{ is available}\}|}{|\{y \mid y \text{ is in the selected range}\}|}.$$

Structural employment shift is the latest available employment share minus the earliest available employment share:

$$shift(c, s) = v(c, y_{last}, s) - v(c, y_{first}, s).$$

3.2 Data Abstraction

The dataset is a static multidimensional table. The main item is a country–year observation indexed by country and year. Countries are treated as categorical items, not as spatial geometry, because the current tasks prioritize attribute comparison and trajectories over geographic adjacency.

3.3 Domain Tasks

The domain tasks are organized by analytical perspective so that cross-country comparison, single-country temporal inspection, structural-change comparison, and reliability checking are not mixed into one flat list. This organization also separates the main analytical tasks from the supporting data-quality task: coverage inspection helps users judge whether a pattern is trustworthy, but it is not itself the substantive tourism-development question.

3.4 Task Abstraction

The task abstraction translates each domain task into one Munzner-style action–target pair. Keeping each row to a single action and target makes the design mapping clearer: scatterplots support discovering correlations, region hue supports comparing distributions, ternary trajectories support identifying trends, dotplots support comparing trends, and the coverage matrix summarizes data availability.

Table 1: Data abstraction of the main attributes used in the dashboard.

Attribute	Role	Abstract type
Country	Item identifier and part of composite key	Categorical key
Year	Temporal index and part of composite key	Ordered key
Region	Grouping attribute	Categorical attribute
Tourism receipts as % of exports	Tourism-dependence value	Quantitative sequential attribute
Employment in services, industry, agriculture	Three-sector employment composition	Quantitative part-to-whole composition
Employment group	Selector for male/female/gender-average variants	Categorical attribute
Development indicators	GDP, urbanization, poverty, and inequality outcomes	Quantitative sequential attributes
Data mode	Exact-year or latest-available value status	Categorical attribute
Data coverage	Reliability value per country and indicator	Quantitative sequential attribute
Structural employment shift	First-to-latest sector-share change	Quantitative diverging attribute

Table 2: Domain tasks organized by analytical perspective.

Perspective	Domain task	Example question
Cross-country association	T1: Compare tourism dependence with employment-sector composition.	Do high-tourism countries also show higher service employment?
Development comparison	T2: Compare tourism dependence with development indicators.	Are tourism-dependent countries different in GDP, poverty, urbanization, or inequality?
Regional comparison	T3: Compare patterns across regions.	Do countries from the same region cluster?
Temporal pathway	T4: Follow one country’s employment composition over time.	Does a selected country move toward services?
Structural change	T5: Compare first-to-latest employment-sector changes across countries.	Which countries show the strongest sectoral shifts?
Reliability support	S1: Check data availability before interpreting a comparison.	Is a visible pattern based on sufficient coverage?

Table 3: Task abstraction using one action–target pair per task.

Task	Action	Target
T1	Discover	Correlation
T2	Discover	Correlation
T3	Compare	Distribution
T4	Identify	Trend
T5	Compare	Trend
S1	Summarize	Feature: data availability

4 FINAL DESIGN: TOURISM STRUCTURAL EXPLORER

The final dashboard layout is shown in Fig. 1.

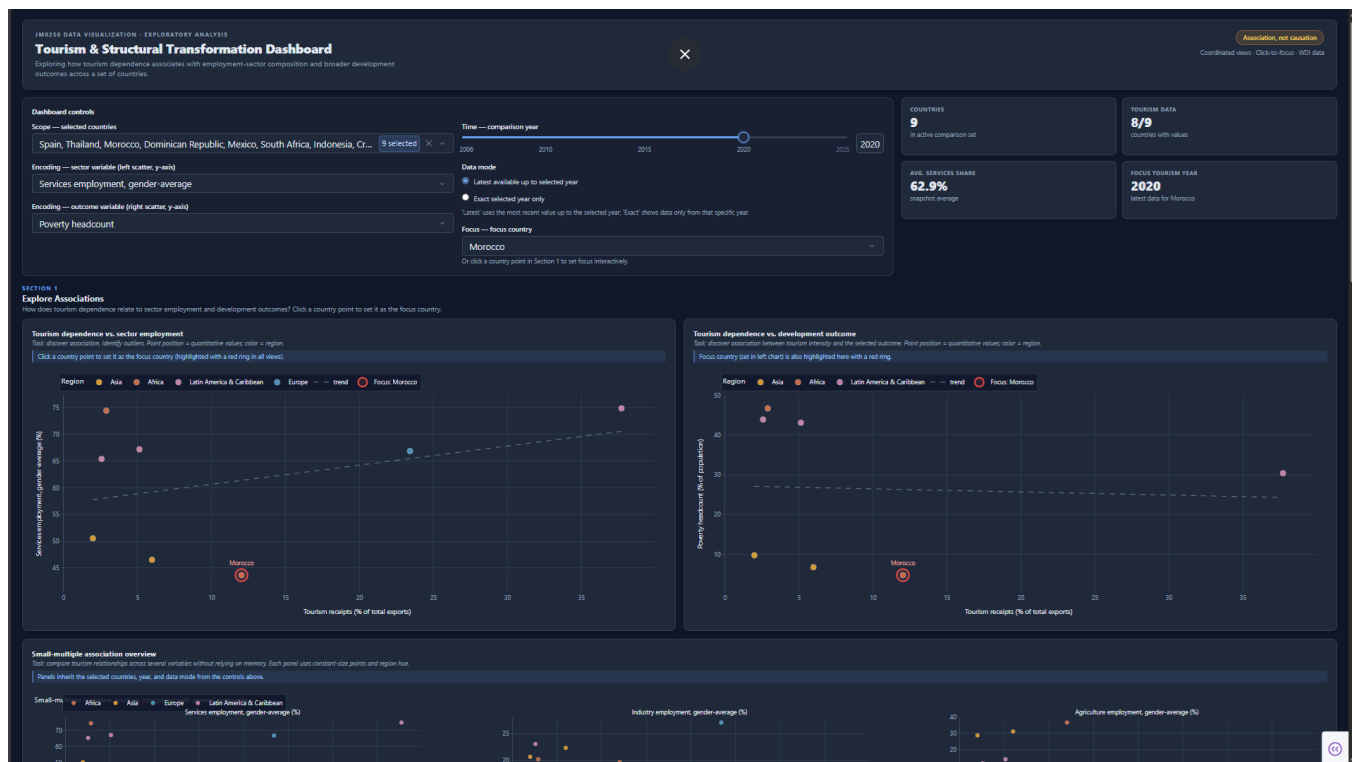


Figure 1: Overview of the *Tourism Structural Explorer*. The dashboard combines global controls, association views, a small-multiple overview, ternary composition views, a structural-shift summary, and data-coverage inspection.

4.1 Association Views

The two main association scatterplots, shown in Fig. 2, compare tourism receipts with a selected employment-sector variable and a selected development outcome. Each point represents a country. Horizontal and vertical position encode quantitative values, which is effective for comparing magnitude and discovering associations [3]. Hue encodes region because region is categorical, and fixed color mappings keep categories stable under filtering. Marker size is kept constant to avoid implying a third quantitative attribute; GDP per capita and service employment values are available in hover when useful as context.

The small-multiple association overview in Fig. 3 addresses the feedback that users otherwise have to remember one tourism-variable relationship while changing dropdowns. Each panel uses tourism receipts on the horizontal axis and a different employment or development variable on the vertical axis. Constant-size marks and shared region hue keep the encoding consistent across panels. The view supports T1 and T2 by making multiple relationships comparable side by side.

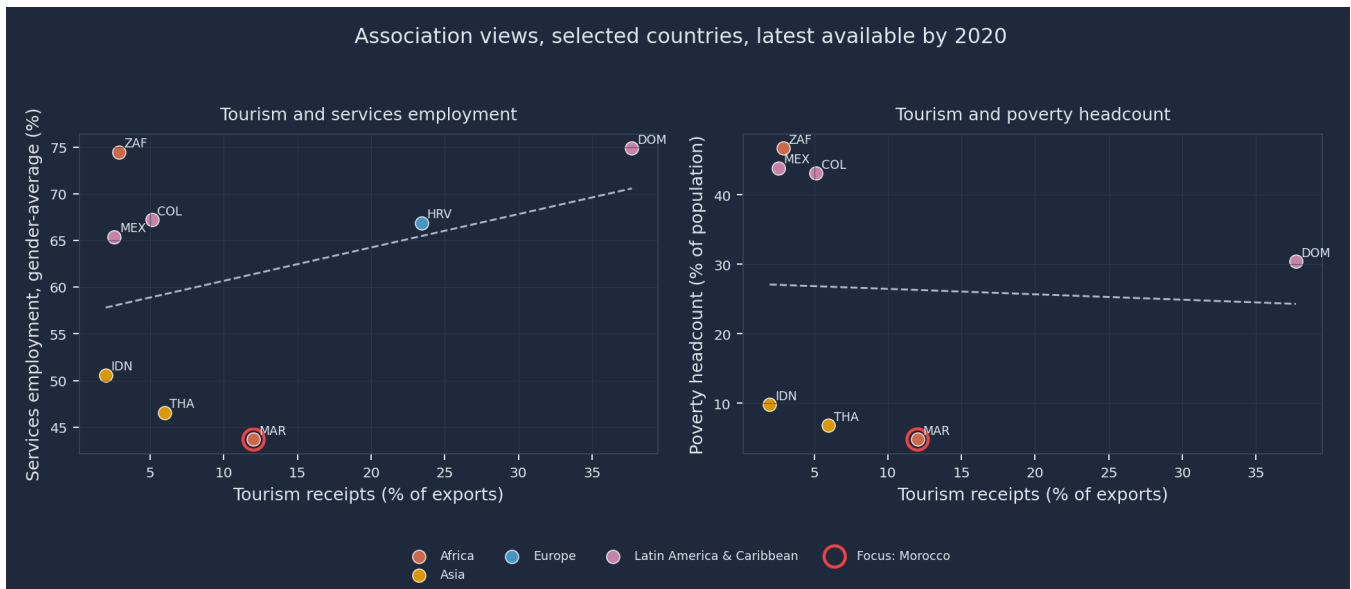


Figure 2: Association views for cross-country comparison. Each point represents a country, position encodes tourism dependence and the selected outcome, hue encodes region, and marker size is constant.

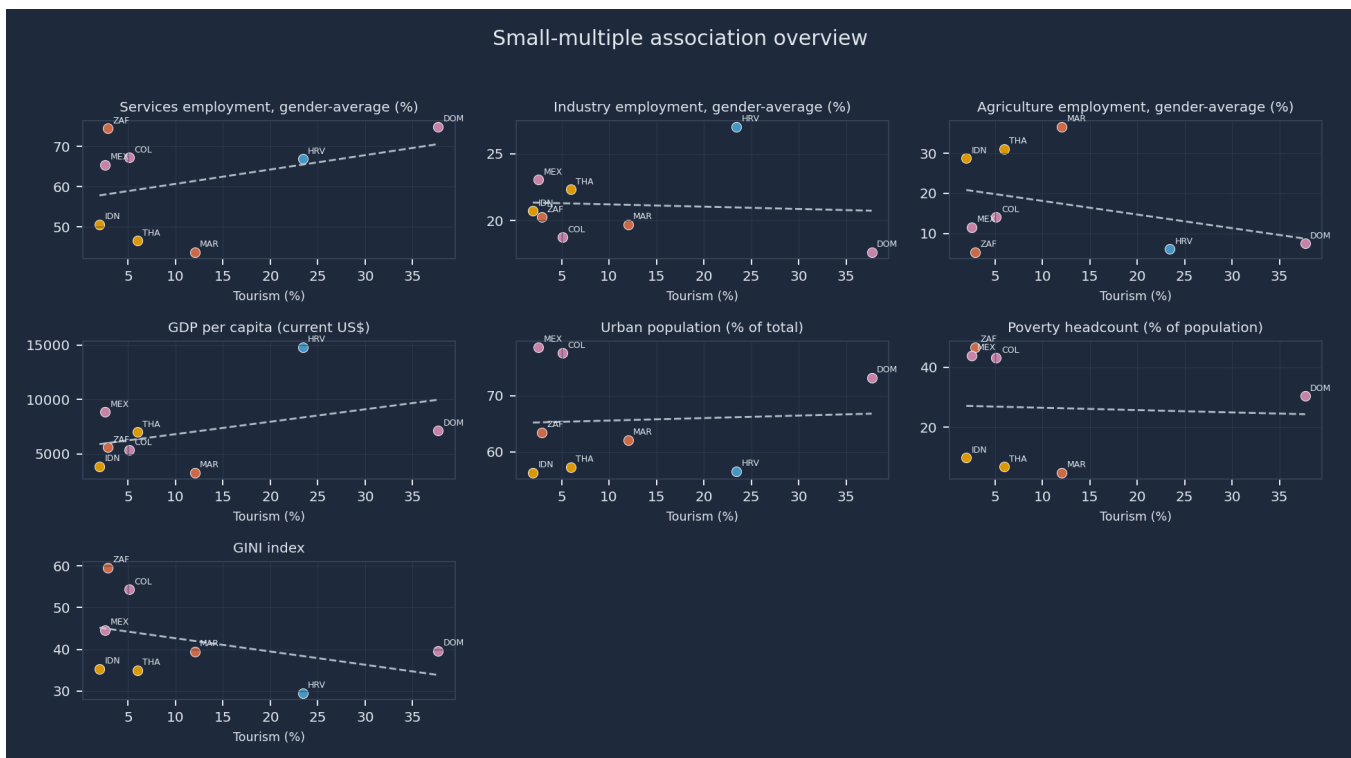


Figure 3: Small-multiple association overview. Each panel compares tourism dependence with a different employment or development attribute using the selected countries, selected year, and data mode.

4.2 Composition and Pathway Views

The ternary views show employment composition, not tourism directly. Services, industry, and agriculture form a constrained three-part composition that sums to approximately 100%, so the ternary plot uses the three sector shares as position. Axis labels and hover details report the three percentages. The snapshot ternary view sup-

ports comparison of selected countries in one compositional space; the focus-country trajectory in Fig. 4 supports T4 by showing ordered change over time. Tourism is linked through the focus-country workflow: users identify a country in the tourism scatterplots and then inspect its employment pathway.

Ternary plots are less familiar and less precise than Cartesian plots, so the dashboard provides hover values and a structural-shift

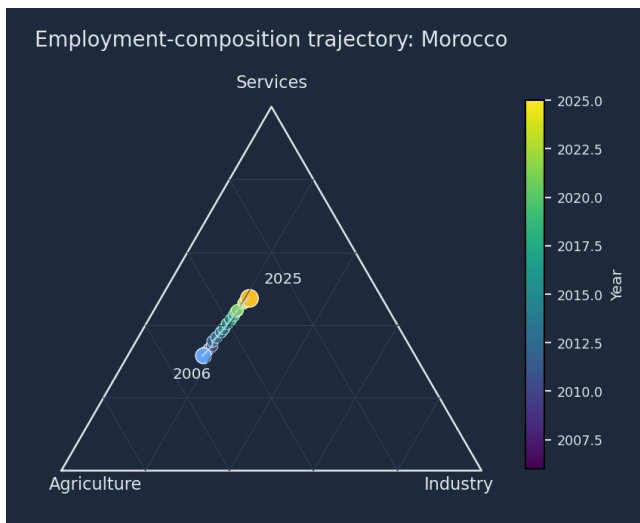


Figure 4: Employment-composition trajectory for Morocco. Each point represents a country-year observation positioned by services, industry, and agriculture shares.

companion view. Alternatives include separate line charts, which make individual sector trends easier to read but do not show the part-to-whole balance, and streamgraphs or stacked areas, which can imply accumulated totals and become hard to compare precisely.

4.3 Structural Shift and Data Coverage

The structural-shift dotplot in Fig. 5 summarizes first-to-latest employment-sector changes for each selected country. A shared zero baseline supports comparison of signed percentage-point changes. This is preferable to grouped or stacked bars for T5 because stacked bars imply part-to-whole totals and grouped bars create many unaligned baselines when several countries and sectors are shown.

The data coverage matrix in Fig. 6 supports S1 rather than serving as a core analytical outcome. It uses the selected countries and prioritizes the currently selected sector and outcome alongside the core indicators. Rows represent indicators, columns represent countries, and color encodes coverage percentage. This helps users decide whether an apparent tourism-development pattern should be interpreted cautiously.

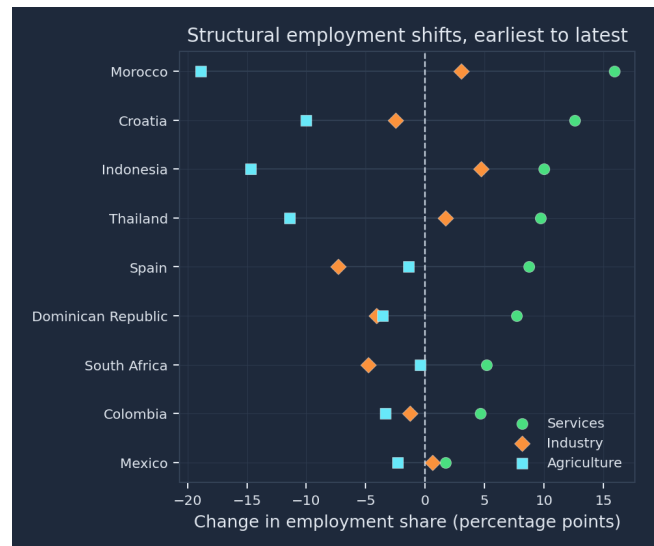


Figure 5: Structural employment shift by country and sector. Each mark shows the percentage-point change between earliest and latest available observations.

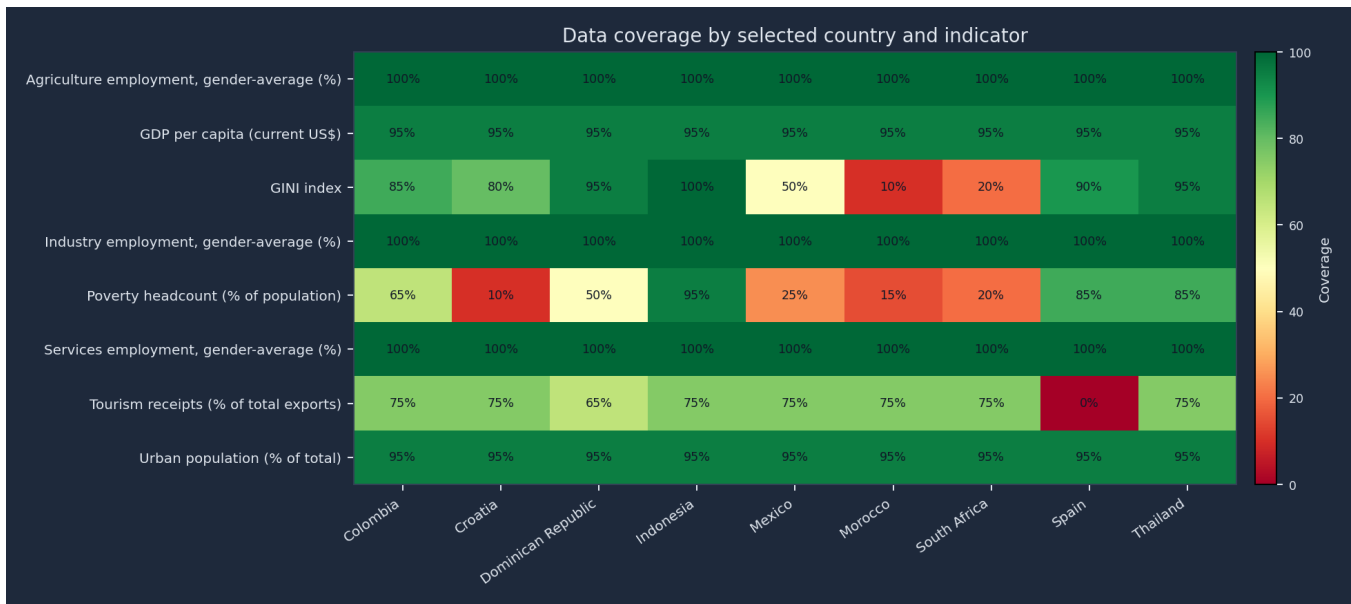


Figure 6: Data coverage matrix for the selected countries and indicators. Rows represent indicators, columns represent countries, and color encodes the percentage of years with available observations.

4.4 Interaction and Coordinated Views

The interaction design follows the overview-filter-details pattern [9]. Users filter by country set, year, data mode, employment sector, and development outcome. Dropdowns change encodings, click selection sets a focus country, linked views update the trajectory and highlights, and hover provides exact values. In Yi et al.'s vocabulary [15], the system supports filter, select, connect, encode, and abstract/elaborate interactions.

4.5 Alternatives and Design Trade-offs

A choropleth map could reveal broad geographic patterns and regional clustering, so it is not inherently inappropriate. It was not chosen as a primary view because the main tasks require quantitative comparison, outlier inspection, variable switching, and temporal pathway analysis rather than adjacency or spatial diffusion. A map could be useful as a future overview if it remains linked to the existing comparison views.

The trend lines in the scatterplots are exploratory summaries only. They help users see broad direction but are not statistical evidence of causality. Overplotting is limited by country filtering and the small project country set, but larger datasets would require stronger aggregation, transparency, search, or faceting.

5 USE CASES AND EVALUATION

The system is evaluated through task-based use cases rather than a formal user study. Each use case starts from a domain question and records the interaction path that supports the analysis.

5.1 Use Case 1: Tourism Dependence and Employment Pathway

Question. Do tourism-dependent countries show a more service-oriented employment structure, and does a selected country move toward services over time?

Interaction path. The user selects the 2020 comparison year and latest-available data mode, inspects the tourism-services association view and the small-multiple overview, selects Morocco as the focus country, and then compares the ternary trajectory with the structural-shift dotplot.

Observation. The 2020 latest-available association view shows that the relationship between tourism dependence and service employment is not uniform across the selected countries. The Dominican Republic is the clearest high-tourism and high-services case, with tourism receipts equal to 37.7% of exports using the latest available value from 2018, and services employment equal to 74.9% in 2020. Croatia is also tourism-dependent, with tourism receipts equal to 23.4% in 2020, but has a lower services share of 66.9%. Morocco has tourism receipts of 12.0% in 2020, but only 43.7% services employment. Selecting Morocco reveals why a single scatterplot is insufficient: its employment structure changes strongly over time. Between 2006 and 2025, Morocco's services share increases from 31.6% to 47.5%, while agriculture decreases from 50.6% to 31.6%. The structural-shift dotplot summarizes this as a +15.9 percentage-point shift in services and a -18.9 percentage-point shift in agriculture.

Interpretation. This use case shows that tourism dependence and service employment cannot be interpreted as a simple one-dimensional relationship. Morocco is not among the highest-service countries in the selected-year scatterplot, but the trajectory and dotplot reveal one of the strongest service-oriented structural shifts in the selected country set. The coordinated workflow is therefore useful because the association view identifies cross-country position, while the ternary trajectory and dotplot explain temporal change.

Limitations. The observation is exploratory and does not show that tourism caused Morocco's employment shift. Other factors such as urbanization, industrial policy, education, trade structure, or broader economic development may also contribute. The tourism indicator also captures international tourism receipts relative to exports, not domestic tourism or indirect tourism employment.

5.2 Use Case 2: Development Indicators and Data Reliability

Question. Does tourism dependence appear related to poverty or inequality, and is the comparison reliable enough to interpret?

Interaction path. The user selects poverty headcount and then Gini index as the development outcome, inspects the tourism-development association view, compares the exact-year and latest-available modes, and checks the data coverage matrix for the selected

countries and indicators.

Observation. The tourism–poverty and tourism–inequality views do not support a simple relationship. For example, the Dominican Republic has the highest tourism dependence in the selected set, with tourism receipts of 37.7% of exports using the 2018 latest-available value, but its poverty headcount is 30.4% in 2020. South Africa has much lower tourism dependence, 2.9% in 2020, but much higher poverty, 46.7%, using the latest available poverty value from 2014. Morocco has tourism receipts of 12.0% in 2020 and a poverty value of 4.8%, but that poverty value is from 2013. Croatia has high tourism dependence at 23.4% in 2020, but no latest-available poverty value in the selected snapshot.

Reliability check. The coverage matrix changes how the association view should be read. Employment-sector indicators have complete coverage for all selected countries, and GDP per capita and urban population have 95% coverage. In contrast, poverty coverage is much weaker: Morocco has 15%, South Africa 20%, Mexico 25%, and Croatia 10%. Gini coverage is also uneven, especially for Morocco at 10% and South Africa at 20%. Therefore, poverty and inequality comparisons are less reliable than employment-structure comparisons.

Interpretation. This use case demonstrates the role of the coverage matrix as a supporting reliability view. The scatterplot can show apparent tourism–development patterns, but the coverage matrix prevents users from treating sparse or old observations as equally reliable as complete employment or GDP series.

Limitations. The poverty and inequality views are useful for hypothesis generation, but they should not be used as evidence of a robust relationship without further statistical analysis and better temporal coverage. The dashboard deliberately exposes this limitation instead of hiding missingness.

6 IMPLEMENTATION AND REPRODUCIBILITY

The dashboard is implemented in Python using Dash [6], Plotly [7], pandas [10], and NumPy [4]. The standard entry point is `python app/tourism_structural_dashboard_designed.py`; the root-level script remains the main implementation. The app expects `employmentdata.csv` and `tourism15indicators.csv` in the repository root. Full setup commands are documented in `README.md`.

7 CHANGES AFTER INTERIM FEEDBACK

The design and report were revised after interim feedback in the following ways:

- **Data abstraction:** raw WDI variables are separated from project-derived attributes, and formulas define gender-average employment, latest-available values, data coverage, and structural shift.
- **Task analysis:** domain tasks are grouped by analytical perspective, and reliability is framed as a supporting validation task.
- **Task abstraction:** each abstract task uses one action–target pair.
- **Visual encoding:** variable marker-size encodings were removed, while stable region and sector palettes were retained.
- **Variable comparison:** a small-multiple association overview was added to reduce the memory burden of comparing several tourism relationships.
- **Ternary and coverage views:** ternary axes now use percentage labels, tourism is described as linked through focus-country selection, and the coverage matrix remains coordinated with selected countries and indicators.

8 CONCLUSION AND FUTURE WORK

Tourism Structural Explorer addresses the project goal by combining cross-country association views, employment-composition trajectories, structural-shift summaries, and data-coverage inspection in one coordinated dashboard. The final design supports exploratory analysis rather than causal inference: users can compare countries, identify outliers, inspect focus-country employment pathways, and judge whether visible patterns are supported by sufficient data coverage.

The use cases show that the coordinated workflow is useful because no single view answers the full question. Scatterplots reveal association patterns and outliers, the small-multiple overview reduces memory-based comparison across variables, the ternary trajectory explains how one country’s employment structure changes over time, the dotplot summarizes sectoral shifts across countries, and the coverage matrix contextualizes reliability. The main limitation is that the dashboard uses one tourism-dependence indicator and does not model causal mechanisms or external confounders such as education, trade structure, industrial policy, or broader macroeconomic development. Future work could add a linked choropleth overview for geographic context, support richer country grouping, include additional tourism indicators, and evaluate the dashboard with representative users.

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